

Data Protection & Do-No-Harm Assessment

AegisAid — Privacy-Safe Humanitarian Resource AI | WiePrompt Hackathon: Data for the Displaced

Humanitarian organizations operate under the ethical imperative to **Do No Harm**. In digital resource allocation systems, the very data collected to direct aid (food insecurity, medical conditions, household structure) can become a liability if exposed — enabling targeting, harassment, deportation, or exploitation. This assessment evaluates AegisAid's threat landscape and the technical safety guarantees implemented to protect the ~120,000 displaced individuals across the three partner camps.

Part 1 — Threat Model & Risk Analysis

Risk Vector	Threat Description	Harm Potential
1. Linkage Attacks (Identity Disclosure)	Attackers cross-reference anonymized aid data with external registries (e.g., national ID, electoral rolls) to re-identify individuals.	Arrest, harassment, deportation, or targeted violence against specific individuals.
2. Reconstruction Attacks (Attribute Disclosure)	Repeated queries against aggregate statistics (e.g., "total diabetes cases in Camp Alpha") to infer individual attributes.	Exposure of sensitive health or demographic information, leading to discrimination or targeted aid denial.
3. Coercive Administrative Data Demands	Host-government or coalition partners pressure operators to hand over the active database for targeted operations.	Loss of the active database for set targeted deportation or logistical tracking.

Part 2 — Mitigation Blueprint & Technical Guarantees

To be **structurally incapable** of exposing identities, AegisAid layers three independent defenses: **Layer 1 (k-Anonymity)** suppresses any reporting group smaller than k → **Layer 2 (Local Differential Privacy)** adds calibrated Laplace noise to every aggregate before it leaves the device → **Layer 3 (Offline-First)** runs the whole pipeline client-side with no central database to breach.

1. Local Differential Privacy (Laplace Mechanism)

$M(x) = f(x) + Y$, where $Y \sim \text{Laplace}(0, \Delta f/\epsilon)$. Sensitivity $\Delta f = 1.0$ (max influence of one household on a normalized 0–1 score). At $\epsilon = 0.5$ for shared aggregates, outputs from any two datasets differing by a single household are near-indistinguishable, defeating reconstruction attacks with high statistical confidence.

2. Structural Guarantee: k-Anonymity ($k = 15$)

Records are grouped by quasi-identifiers (camp, gender, age band); any group of fewer than 15 people is suppressed or generalized before export. On the reference 10,000-record synthetic dataset (seed = 42, reproducible via scripts/synthetic_gen.py) this suppresses 20 records (0.2%), all from the smallest "Other gender × 65+" cohort per camp — no exported statistic can ever be traced to a group smaller than 15.

3. Decentralized Architecture: Client-Side "Offline-First"

The dashboard runs entirely in-browser on local devices with zero central database dependency. If a field worker's device is seized or the local network is intercepted, there is no centralized store to breach; raw data is never transmitted across borders or written to disk unencrypted.

4. Agency & Dignity: Inclusive Consent

A visual + spoken consent panel (English, Arabic, French, Somali) supports low-literacy intake. Declining consent does not deny aid — the household instead receives the camp-average resource basket, removing any incentive to coerce participation.

Conclusion & Trade-offs

Privacy protection is not free: adding DP noise and suppressing small k-anonymity classes reduces allocation precision by approximately **8%** versus an unprotected optimum. We accept this utility cost as the price of guaranteeing the physical safety of displaced persons, directly satisfying the brief's requirement that trade-offs be **made explicit, not just claimed** — while still advancing SDGs 1, 2, 3 and 10.